UNRAVELLING THE MECHANISMS OF MANIPULATING NUMBERS IN LANGUAGE MODELS

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ABSTRACT

Recent work has shown that different large language models (LLMs) converge to similar and accurate input embedding representations for numbers. These findings conflict with the documented propensity of LLMs to produce erroneous outputs when dealing with numeric information. In this work, we aim to explain this conflict by exploring how language models *manipulate* numbers and quantify the lower bounds of accuracy of these mechanisms. We find that despite surfacing errors, different language models learn interchangeable representations of numbers that are systematic, highly accurate and universal across their hidden states and the types of input contexts. This allows us to create universal probes for each LLM and to trace information — including the causes of output errors — to specific layers. Our results lay a fundamental understanding of how pre-trained LLMs manipulate numbers and outline a potential of more accurate probing techniques in addressed refinements of LLMs' architectures.

1 INTRODUCTION

One major limitation of large language models (LLMs) is their inability to guarantee factually precise and correct outputs. This is a fundamental trait due to the probabilistic nature of transformer models, and thus not a solvable problem, as has recently been emphasized by (Kalai et al., 2025). However, knowing the extent of the problem (and whether it can be mitigated) is a feasible task; one that is important given the recent ubiquity of LLMs in research and broader society.

Mathematical problems provide an ideal playground in which to assess the current ability of LLMs to manipulate precise information. Because it operates through a set of formal rules and produces computationally verifiable outputs, mathematics provides the opportunity to rate not only the tendency of a model to output incoherent information (Sun et al., 2024), but also to examine in detail the coherence of the reasoning steps outputted by a given LLM. While this research direction is rife with challenges, ranging from manipulating implicitly visual data in geometry problems (Ahn et al., 2024) to mapping plain language onto specific types of problems (Duchnowski et al., 2025), it has also been met by genuine successes, e.g., as an aid to solve mathematical problems (Li et al., 2025).

A large body of work has highlighted limitations of LLMs when it comes to arithmetic tasks (Xu et al., 2025; Kvinge et al., 2025; Satpute et al., 2024; Lee et al., 2025; Bertolazzi et al., 2025; Nikankin et al., 2025), which in turn has led to various proposals to alleviate these limitations centered on re-designing numeric representations to be principled and precise (e.g., Charton, 2022; Feng et al., 2024; Golkar et al., 2023). More recent developments have suggested that the issues models

encounter with arithmetic need not be tied to the quality of their representations of numbers. Several works have attempted to characterize the types of representations models rely on to perform arithmetic computations (Wennberg & Henter, 2024; Zhou et al., 2024a; Zhu et al., 2025; Kadlčík et al., 2025). This research direction has proposed that number representations stand out in terms of their structure, with Zhou et al. (2024a) and Kadlčík et al. (2025) arguing that models rely on sinusoidal structures, whereas Zhu et al. (2025) suggest that representations are better described using a linear model than a multi-layer perceptron. These results have also been combined with observations as to how models process and succeed at arithmetic tasks.

In this paper, we provide evidence that LLMs converge to systematic and accurate *sinusoidal* representations that are maintained throughout the model in different forms. These representations are largely equivalent across LLMs of different sizes and families. Building upon this, we find that numeric information from LLMs can be accurately extracted using universal probes and show that more accurate numeric probes allow us to attribute up to 94% of model errors in arithmetical reasoning to particular layers, overriding the existing, correct results.[1]

2 Related work

Math capabilities of LLMs The seminal remarks of Brown et al. (2020) that LLMs that are large enough develop arithmetic capabilities have ushered in a large body of work aimed at evaluating the mathematical capabilities of LLMs (Hendrycks et al., 2021; Cobbe et al., 2021; Sun et al., 2024; Yu et al., 2024) or lack thereof (Nikankin et al., 2025). This in turn has encouraged a focus on how to bolster the capabilities of LLMs or interpret how LLMs perform computations (Zhang et al., 2024; Stolfo et al., 2023). Most relevant to ours, some of the previous work especially focuses on how numbers should be represented in principle. For instance, Charton (2022) assess an impact on models performance in linear algebra problems when employing different encoding schemes based on scientific notation on linear algebra problems. Golkar et al. (2023) propose to encode numeric values by incorporating a scaled, learned control token <NUM>. Feng et al. (2024) remark that the precision of the numeric representation type impacts performance on arithmetic tasks, and argue to use the logarithmic-precision architecture of Feng et al. (2023).

Numeric representations in LLMs The increased focus on evaluating LLM performance in verifiable domains such as math problems has also led to increased interest in the mechanisms by which LLMs compute arithmetic functions. Nanda et al. (2023) demonstrate how a model trained from scratch on modular addition relies on trigonometric operation: it maps inputs onto a unit circle, corresponding to a specific rotation, and then learns to combine the two rotations to derive a valid solution. Kantamneni & Tegmark (2025) apply circuit analysis to highlight how general pretrained models perform addition, and highlight that representations are mapped onto a helix that can be manipulated using trigonometric operations. Zhou et al. (2024b) recast these observations in terms of Fourier components (since Fourier transformations map arbitrary functions unto combinations of periodic trigonometric functions), and highlight the distinct role of attention and feedforward sublayers. Focusing more narrowly on representations rather than processing, Zhu et al. (2025) argue that non-linear probes do not provide a better fit than linear probes. Levy & Geva (2025) remark that it is possible to retrieve the digit (value mod 10) of numeric inputs with high accuracy. Kadlčík et al. (2025) propose a linear probe architecture that factors in the sinusoidal nature of the embeddings of numbers, and show that they can retrieve numeric precision with high accuracy.

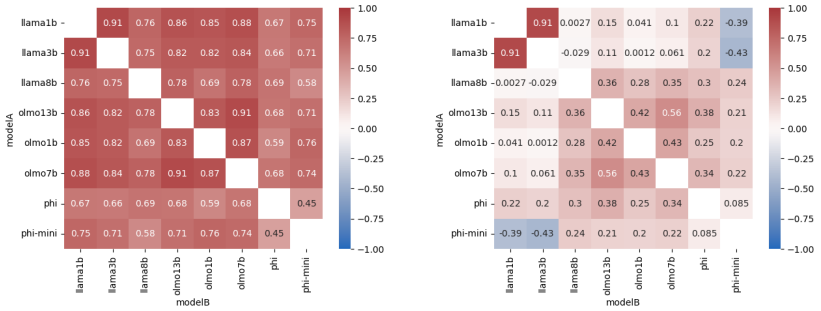
In summary, there is a growing body of evidence stating that models rely on trigonometric representations for numbers. Here, we build upon Kadlčík et al. (2025) specifically, as the high accuracy of their proposed probe provides new opportunities and novel research directions — as we find, also in tracking and verifying the accuracy of a model's *inner* representations throughout its computations, robust across contexts, model types and operations.

[1] We make all our methods and analyses available to any use at the project repository: `https://github.com/prompteus/numllama`

3 UNIVERSALITY OF SINUSOIDAL REPRESENTATIONS OF NUMBERS

3.1 MODELS LEARN EQUIVALENT REPRESENTATIONS OF NUMBERS

The first point we address is the distinctiveness of LLMs' representations of numbers. Results from Kadlčík et al. (2025) and Zhu et al. (2025) suggest that different models converge to the same *type* of number representations, but they do not provide a direct assessment of how *closely* the representations match across models – which could evidence that a shared representation is a causal consequence of architectural bias and optimization process rather than a coincidental artifact. Following Kadlčík et al., we focus on the input embeddings from eight LLMs of diverse sizes and families with open-sourced, pre-trained checkpoints: OLMo 2 (OLMo et al., 2025), Llama 3 (Grattafiori et al., 2024) and Phi 4 (Abdin et al., 2024).



(a) Cosine-based RSA for number tokens (b) Cosine-based RSA for random tokens

Figure 1: Representational similarity analysis (RSA) scores

To quantify whether models converge to similar embeddings, we start by conducting a simple Representational Similarity Analysis (RSA; Kriegeskorte et al., 2008) across the input embeddings for number word-pieces in paired models.[2] As a baseline, we compute RSA scores for a random sample of 1000 pieces present in the vocabularies of all the models under consideration. Results of the analysis are displayed in Figure 1. Across all pairs of models, we find that number embeddings systematically yield higher RSA scores than a random sample of word pieces. While some models (especially in the Phi family) are not as well aligned with other models, we do observe high scores both within and across families, demonstrating that models converge to embedding spaces with equivalent similarity structure.

Another approach to quantifying the similarity of frequential representations characteristic for numbers Kadlčík et al. (2025); Zhou et al. (2024a) is by inspecting their base frequencies through Fourier decompositions. Applying a PCA transformation followed by a Fourier transform allows us to quantify the *magnitude* of individual frequencies (Figure 2a). Subsequently, we determine whether two models agree as to how they rank frequencies by computing a simple intersection-over-union (IoU) of the top k frequencies. We repeat this process for every pair of models, using the input embeddings of number pieces (Figure 2b) as well as that of random pieces as previously (Figure 2c). When considering the top $k = 63$ frequencies,[3] we find **perfect agreement across all models** in terms of number pieces.

In summary, our analyses show that the LLMs from different families learn to represent numbers in a similar topology and trigonometric features using the same dominant frequencies. This is specific to numbers – we do not reproduce these observations with other word pieces shared across models.

[2]RSA is a second-order similarity measurement. Let $A = \{\mathbf{a}_1, \dots \mathbf{a}_n\}$ and $B = \{\mathbf{b}_1, \dots \mathbf{b}_n\}$ be two sets of matching representations, in our case the input embeddings for number word-pieces in two models θ_A and θ_B. We can assess the extent to which A and B encode the same type of similarity structures by simply computing the (Spearman) correlation $\rho_{\mathbf{s}_A, \mathbf{s}_B}$, where $\mathbf{s}_A = (\cos(\mathbf{a}_1, \mathbf{a}_2) \dots \cos(\mathbf{a}_{n-1}, \mathbf{a}_n))$ and $\mathbf{s}_B = (\cos(\mathbf{b}_1, \mathbf{b}_2) \dots \cos(\mathbf{b}_{n-1}, \mathbf{b}_n))$ are two vectors tracking all pairwise similarities within A and B respectively.

[3]See Section A.1 for details how this value of k is found.